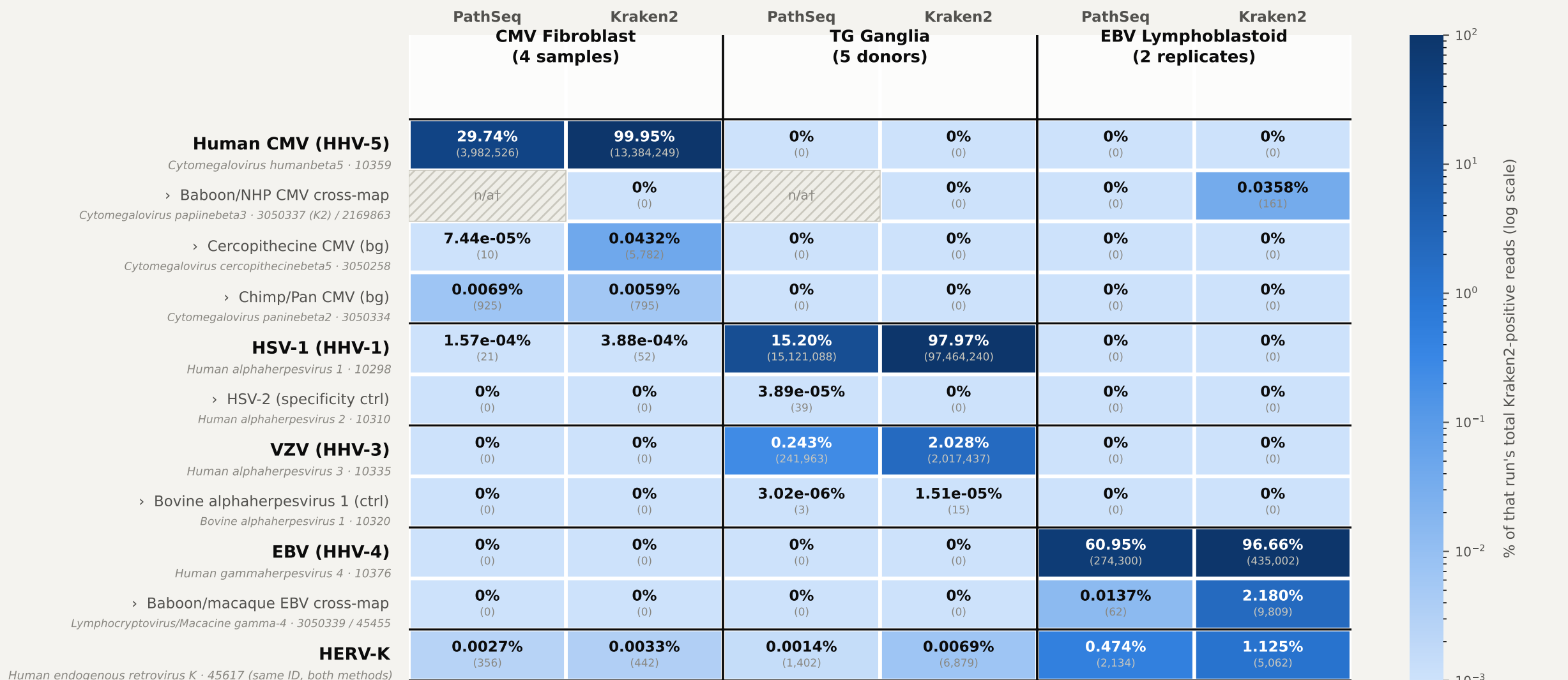


Kraken2 × PathSeq Concordance — High-Resolution

Rows grouped by taxonomic lineage — each main virus (bold) with every related sub-taxon (cross-species artifact, within-genus background, specificity control) indented directly beneath it. Same shared-Kraken2-denominator normalization as the primary figure.



Normalization: for each run, % = (taxon's raw read count ÷ sum of Kraken2 raw reads across every taxon shown for that run) × 100, applied identically to both columns. Every cell is now real, checked data — every taxon (including HSV-2 and Bovine alphaherpesvirus 1, both found via corrected name search since Kraken2 renames these under its current taxonomy vintage) was cross-validated against all 3 cohorts, both tools (2026-08-16). Bovine alphaherpesvirus 1 (taxon 3050243, "Varicellovirus bovinealpha1") has a real positive: 15 reads in one TG donor (TG3) only, zero elsewhere — consistent with an incidental low-level background detection, not a systematic signal. HSV-2 is a confirmed true-negative everywhere. HERV-K (45617): present in all 3 cohorts, as expected for ubiquitous low-level human genomic background — PathSeq uses the SAME taxon ID as Kraken2 here (unlike every herpesvirus, where taxonomy-vintage differences forced different IDs per tool), and its parent/species/K113-subtype nodes report numerically identical counts in every sample, meaning essentially all signal resolves to the K113 subtype specifically. † PathSeq has no taxon distinct from 10359 representing Kraken2's specific proxy-species artifact — see the primary figure's footer for the full explanation of both n/a† cells.
Source: docs/pathseq_validation_results_2026-08-15.md, research/cmv_taxonomy_investigation.md.